

The Psychometric Properties of a Tinnitus Handicap Questionnaire*



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ABSTRACT

The psychometric properties of a tinnitus handicap questionnaire are reported. There were two phases in this study. In Phase I, 87 questions were administered to 100 tinnitus patients. From their responses, 59 items that were either redundant, insensitive, or had low item-total correlations were eliminated. In Phase II, the resulting 27-item questionnaire was administered to 319 patients. Fifty-three of these patients also completed psychological and psychophysical measures that were used to validate the questionnaire. A factor analysis of patients' responses revealed a three-factor structure. These three factors appeared to reflect the physical, emotional, and social consequences of tinnitus (Factor 1), hearing ability of the patient (Factor 2), and the patients' view of tinnitus (Factor 3). Although the 27-item questionnaire had high internal consistency reliability and validity as reflected by correlations with life satisfaction and depression scales, it is recommended that only the items on the Factor 1 and the Factor 2 subscales be scored because of the low internal consistency reliability of the Factor 3 subscale. This questionnaire can be used to compare a patient's tinnitus handicap with the norm, identify specific areas of handicaps, and to monitor a patient's progress with particular treatment programs.

TINNITUS IS A debilitating disorder which can lead to anguish and depression. Chung, Gannon, and Mason (1984) reported that 6 to 17% of the population is affected by tinnitus. The incidence is even higher in the hearing-impaired population, where 70 to 85% of individuals with some forms of ear diseases reported tinnitus (Fowler, 1944; Glasgold & Altmann, 1966; House & Brackmann, 1981). McFadden (1982) and Vernon, Johnson, Schleuning, and Mitchell (1980) also

assert that over 85% of tinnitus patients have some hearing loss. Tinnitus is a multifaceted disorder that can affect an individual's hearing, health, emotion, and life-style. Tyler and Baker (1983) found that the most common problems associated with tinnitus are falling asleep, the persistence of the disorder, understanding speech, despair, annoyance, difficulty with concentration, and drug dependence. Stephens and Hallam (1985) also found that tinnitus sufferers showed an elevation on the anxiety and depression scales of the Crown-Crisp Experiential Index (1979). Because the extent of this disorder is diversified, patient evaluation needs to consider all of the potentially handicapping areas. Unfortunately, no methods are presently available that can quantify reliably the severity of handicap experienced by a tinnitus patient.

Another facet of tinnitus research that deserves more attention is the standardization of measures to validate the efficiency of different treatment approaches. Treatments for tinnitus include medication (Goodey, 1981; Melding, Goodey, & Thorne, 1967), electrical stimulation (Chouard, Meyer, & Maridat 1981; Kuk et al, 1989; Vernon & Fenwick, 1985), biofeedback (House, Miller, & House, 1979), and counseling (Sweetow, 1986). Although each of these approaches achieves varying degrees of success, the criteria used to define success varied. One cannot easily and reliably determine the efficacy of each treatment approach. The use of standardized measures could ensure uniform comparison among studies.

The purpose of the present investigation was to develop a tinnitus handicap questionnaire to measure the patients' perceived degree of handicap due to tinnitus. Because of the high prevalence of hearing loss in tinnitus patients, no attempts were made to restrict this questionnaire to patients with tinnitus only (i.e., no hearing loss). The factor structure of the questionnaire, its reliability, and construct validity were determined.

There were two phases to this investigation. The purpose of the first study was to generate a manageable number of items to be included in the questionnaire for final validation. The purpose of the second study was to determine the psychometric properties (i.e., reliability, validity, and factor structure) of the questionnaire.

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STUDY I

Method

A pool of 87 items was generated based on Tyler and Baker's (1983) report. These items were grouped arbitrarily to reflect the effects of tinnitus on a particular aspect of the individual's well-being. These groups (or subscales) included the effects of tinnitus on hearing, life-style, health, and the emotion of the sufferers. Additional items that addressed others' reactions to the tinnitus sufferers were also included under the subscale others. Twelve items were grouped under hearing. Twenty-seven items were grouped under life-style. Eighteen items were grouped under health. Twenty items were grouped under emotion and ten items were grouped under others. The list of 87 items was administered at the Department of Otolaryngology of the University of Iowa Hospitals and Clinics to 100 adult patients (older than 18 years) who reported tinnitus at the time of the audiological evaluation. This included patients whose primary complaint was tinnitus and those whose tinnitus was secondary to their hearing loss. Twenty-three of the patients were female. The mean age of all the patients was 53 years. Patients were provided a copy of the 87-item questionnaire. Patients were instructed to write a number between "0" and "100" along each item to represent how much they agreed with the specific item. Patients assigned a 0 if they strongly disagreed with the item, and a 100 if they strongly agreed with the item.

Analysis and Results

The purpose of performing a preliminary analysis was to select a small sample of items that could be administered efficiently but that was sensitive to the tinnitus patients' perceived handicap.

Three criteria were used in the item selection process. These criteria were not exclusive of each other, and compromises were made by the investigators in the selection process. One criterion was to eliminate items that were insensitive to differences among subjects. A frequency count of the proportion of time a specific rating was assigned to each item was performed. Items that were assigned a 0 or a 100 over 50% of the time were assumed insensitive and were subsequently deleted. The Appendix contains the 87 items and the proportion of 0 and 100 ratings. An example of this was item Emotion 17, "I have considered suicide because of tinnitus," where 84% of the patients rated 0 and only 2% of the patients rated 100. Although this item may be very crucial in identifying the emotional state of some tinnitus patients, it was not included because this item was unable to differentiate 84% of the patients. Items that were selected but had over 50% 0 rating included items Life style 24, Health 8, and Health 9. These items were retained to meet the other criteria.

The second criterion was to eliminate redundant items. This was achieved by examining the correlation coefficients among the items contained within each subscale. Items that shared high correlations with other items were examined to select representative items that could replace other items. An example was item Hear 9, "Tinnitus has caused a reduction in my speech understanding ability." This item correlated highly with items Hear 1 ($r = 0.63$) Hear 3 ($r = 0.63$), Hear 6 ($r = 0.73$), Hear 7 ($r = 0.68$), Hear 10 ($r = 0.68$), and Hear 11 ($r = 0.62$). These items probably reflected listening in quiet situations. Because of the correlation, these items were replaced by Hear 9. On the other hand, items like Hear 2 (understanding in noise, $r = 0.50$), Hear 4 (localization, $r = 0.55$), and Hear 8 (warning signals, $r = 0.52$) had weaker

correlations with Hear 9. These items appeared to reflect different hearing abilities and were therefore retained. This approach reduced the total number of items, but could have also reduced the internal consistency reliability of the questionnaire. Consequently, a third criterion was necessary to maximize the internal consistency of the test.

The measurement of internal consistency of a test is a way of estimating its reliability. The assumption is that tests with homogeneous items (i.e., internally consistent) are more reliable than tests with heterogeneous items. To estimate internal consistency, item-total correlation coefficients for each item and Cronbach's α were determined (SPSS-PC+, 1988). The item-total correlation reflects the correlation of an item with a total score computed using all the remaining items on the test. Cronbach's α (Cronbach, 1960) theoretically reflects the expected squared correlation between scores on the scale in question and the person's true score on the dimension being assessed (which cannot actually be observed). Thus, Cronbach's α can also be interpreted as representing the proportion of true score variation in a measure, with $(1-\alpha)$ representing the proportion of error in the measure.

Items with high item-total correlation coefficients were selected to maximize the reliability of the final questionnaire. Selected items with low item-total correlation coefficients were Emotion 13 and Others 3. These items were selected based on the first criterion (i.e., sensitivity). The Appendix presents the item-total correlation coefficients. Using these criteria, a total of 27 items were selected from the original 87-item pool. These items had item-total correlation coefficients ranging from 0.14 to 0.78, reflecting a relatively heterogeneous sample of items. The Cronbach's α of the 27-item questionnaire was 0.93.

STUDY II

Method

The 27-item questionnaire was administered to 319 tinnitus patients who reported tinnitus during their audiological evaluation at otolaryngology offices in the state of Iowa, the House Ear Institute, the Iowa City VA Medical Center, the University of North Carolina Hospitals and Clinics, and The University of Iowa Hospitals and Clinics. However, only 275 patients completed the whole questionnaire. Consequently, the normative sample was based on the responses of these 275 patients. The age and gender distributions of these patients are summarized in Figure 1. The mean age of the male

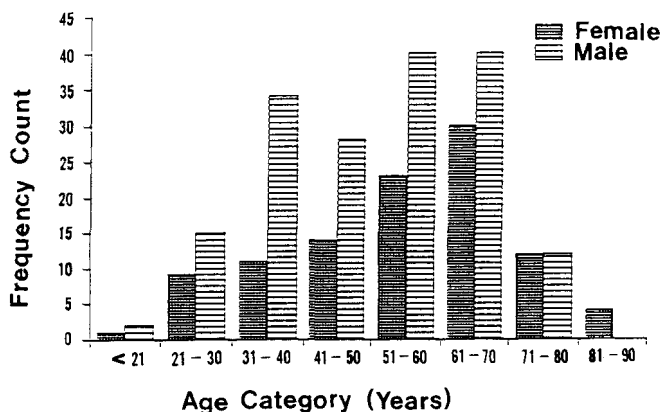


Figure 1. Age distribution of the 275 patients who completed the full 27-item tinnitus handicap questionnaire.

patients was 50.0 years and the mean age of the female patients was 55.5 years. Patients rated each item on the same 0 to 100 scale following the same instructions used in Study I. For 53 patients seen at the Iowa City VA Medical Center and the University of Iowa Hospitals and Clinics, additional psychological scales and psychophysical measures were administered for validation purposes. Thirty-three of these patients were male and their mean age was 52 years. The mean age of the female patients was 54 years. The psychological scales included:

Life Satisfaction This scale consists of a set of 10 items selected from Lohmann's (1980) report. Patients were instructed to use a 4-point scale to indicate their degree of satisfaction with their lives. An average score of 4 suggested that the patient was satisfied with life all of the time whereas an average score of 1 suggested that the patient was never or rarely satisfied with life. Cutrona, Russell, and Rose (1986) found that the scale was reliable (coefficient $\alpha = 0.79$). If the tinnitus questionnaire is a valid measure of handicap, one may expect a high negative correlation between this scale and the tinnitus handicap questionnaire.

Depression The Zung Self-Rating Depression Scale (Zung, 1965) is an 18-item scale designed to measure the affective, psychological, and physiological symptoms of depression. Like the life satisfaction scale, patients used a 4-point scale to indicate their degree of agreement with the items. An average score of 1 suggested that the patients were never or rarely depressed, whereas an average score of 4 suggested that the patient was depressed all of the time. McGarvey, Gallagher, Thompson, and Zelinski (1982) reported that the scale was a reliable measure when used among elderly adults (coefficient $\alpha = 0.59-0.76$). One may expect high correlations between this scale and the tinnitus questionnaire.

Physical Health Status Several measures of patients' physical health status adapted from the Duke-UNC Health Profile (Parkerson et al, 1981) were administered. First, two subscales from the original Profile consisting of a symptom checklist and a functional abilities measure were used. The scores on these two measures were combined to form scores on the Health Status scale. The symptom checklist assessed the presence and severity of 24 physical symptoms, whereas the functional abilities measure assessed the person's ability to perform daily activities. Patients were instructed to use a 3-point scale to indicate their difficulties. An average score of 3 suggested that the patients had a lot of physical and functional difficulties, whereas an average score of 1 suggested that the patients had no physical or functional difficulties. Reliability of these measures ranged between 0.68 to 0.82 (Cutrona et al, 1986). A third subscale, an Illness checklist, was administered to determine the number of illnesses the patients had experienced in the previous 6 months. A fourth subscale that contained a set of three items (General Health scale) requiring a subjective rating (on a 5-point scale) of (1) one's own health in general, (2) one's health compared to others, and (3) one's health compared to 6 months ago from the Duke-UNC Health Profile was administered. An average rating of 5 suggested that the patient viewed his/her health as much better than others or much better now than 6 months ago, whereas a rating of 1 would indicate otherwise. Moderate reliability was found for a composite score across these three items (coefficient $\alpha = 0.67$).

The correlations between scores on the tinnitus questionnaire and the different physical status scales may be difficult to predict. If tinnitus is a symptom of other illnesses, one may

expect to find high correlations between scores on the questionnaire and scores on the physical scales. On the other hand, if tinnitus is not a symptom of other illnesses, correlations among these scales may not be significant.

Social Desirability The Marlowe-Crowne Social Desirability Inventory was administered to identify individuals who describe themselves in socially desirable terms in order to achieve other's approval. Patients had to answer true or false to each of the 33 items. A socially desirable response was given a score of 1, whereas an undesirable response was given a score of 0. This scale has been standardized on different populations (Crowne & Marlowe, 1964), with reportedly high internal consistency (coefficient $\alpha = 0.88$). Ideally, one would find little or no correlation between scores on this scale and the tinnitus questionnaire, because it is reasonable to assume that tinnitus patients desire as much social approval as individuals with no tinnitus. Its inclusion in the battery of tests was to ensure that scores on the tinnitus questionnaire were not influenced by social desirability.

Psychophysical Measures The patients' audiometric thresholds from 250 to 8000 Hz at octave intervals were first determined. The mean threshold at these frequencies was identified as the mean hearing loss (in dB HL). In addition, three psychophysical tinnitus measures were included. The first measure required patients to match the loudness of their tinnitus to a 500-Hz sinusoid in the ear with the louder tinnitus. That ear was defined as the test ear. In patients with no dominant ear, the test ear was selected arbitrarily. A bracketing approach with a 2 dB step was used. Tinnitus loudness in dB SL was then converted to tinnitus loudness in sones using an equation described by Tyler and Conrad-Armes (1983). This transformation was necessary to account for the difference in hearing thresholds among patients. The second measure required patients to report when the level of a speech-shaped broadband noise was sufficiently intense to just mask the tinnitus in the test ear (minimal masking level, MML in dB SL). An ascending approach was used. The masking noise was not sufficiently intense to mask the tinnitus in seven patients. The third measure required patients to subjectively judge the general loudness of their tinnitus in their daily lives using a 0 to 100 point scale, with 0 meaning "no tinnitus" and 100 meaning "extremely loud tinnitus." These psychophysical measures were used to test the assumption that measured tinnitus loudness would reflect the patients' perceived degree of handicap due to tinnitus. High correlations should be found between the tinnitus handicap questionnaire and these psychophysical measures if this assumption is valid.

Analysis and Results

The mean total score for each subject was first computed by adding their responses to each item. Because items 25 and 26 were written so that a larger number would represent less tinnitus handicap, responses to these items were first inverted by subtracting them from 100 prior to the computation. For the male patients ($n = 171$), the individual mean total score ranged from 0 to 90, with a group mean total score of 35.6 (standard error = 1.7). For the female patients ($n = 104$), the individual mean total score also covered the same 0 to 90 range, but with a mean group total of 39.0 (standard error = 2.5). This difference, however, was not statistically significant ($F[1, 273] = 2.5, p > 0.05$). Because there is no reported literature on the effect of gender on perceived tinnitus handicap, and in view of the lack of statistical significance in total

scores between male and female patients, responses for male and female patients were pooled together. Furthermore, a Pearson product-moment correlation coefficient of 0.09 was obtained between the age of the patients and the mean total tinnitus handicap scores, suggesting that age had negligible effect on perceived tinnitus handicap. For this reason, the responses from patients of all ages (and of both sexes) were pooled for the following statistical analysis.

Factor Structure

The first step was to examine the factor structure of the questionnaire using factor analysis techniques. Factor analysis assumes that correlations between responses on a set of variables (i.e., items on the questionnaire) reflect underlying dimensions or factors. Consequently, one prerequisite for performing a factor analysis is that the observed variables be substantially intercorrelated. This was determined using the Kaiser-Meyer-Olkin (KMO) (SPSS-PC+, 1988) measure of sampling adequacy. The KMO determines the ratio of the magnitude of the observed correlations among variables to the magnitude of the sum of the observed correlations and partial correlations among the variables. A ratio close to 1 would indicate that factor analysis is appropriate on the observed variables. A ratio close to 0 would indicate otherwise.

A KMO ratio of 0.94 was obtained for the set of 27 items, indicating that factor analysis was appropriate to explore the underlying dimensions or factors that governed responses to the questionnaire.

Factor analysis using the method of principal components extraction was performed. Three factors with eigenvalues greater than one were identified. The eigenvalue is a measure of the variance in the items accounted for by a given factor of dimension. A common practice is to assume that factors with eigenvalues greater than 1.0 are meaningful or reliable. The first factor explained 42.6% of the total variance (eigenvalue = 11.50); the second factor explained 9.4% of the total variance (eigenvalue = 2.54); and the third factor explained 5.6% of the total variance (eigenvalue = 1.51). The communality of each variable, or the proportion of variance in each item explained by these three factors, ranged from 0.27 to 0.77. These statistics can be found in Table 1.

Because it was assumed that these three factors might be intercorrelated, an oblique rotation ($\delta = 0$) was performed to transform the factor loading matrix into a more interpretable form. Table 2 presents the factor pattern matrix after the oblique rotation, as well as the correlation coefficients among the three factors. As expected, Factors 1 and 2 were substantially correlated, but the correlation of these factors with Factor 3 was small.

Table 1. Results from the factor analysis using principal components extraction. The communality is the sum of the squared correlations between a variable and each of the three factors.

Item	Communality
1. I do not enjoy life because of tinnitus.	0.52
2. My tinnitus has gotten worse over the years.	0.27
3. Tinnitus interferes with my ability to tell where sounds are coming from.	0.62
4. I am unable to follow a conversation during meetings because of tinnitus.	0.67
5. Tinnitus causes me to avoid noisy situations.	0.34
6. Tinnitus interferes with my speech understanding when talking with someone in a noisy room.	0.67
7. I feel uneasy in social situations because of tinnitus.	0.64
8. The general public does not know about the devastating nature of tinnitus.	0.43
9. I cannot concentrate because of tinnitus.	0.54
10. Tinnitus creates family problems.	0.49
11. Tinnitus causes me to feel depressed.	0.71
12. I find it difficult to explain what tinnitus is to others.	0.37
13. Tinnitus causes stress.	0.77
14. I am unable to relax because of tinnitus.	0.70
15. I complain more because of tinnitus.	0.71
16. I have trouble falling asleep at night because of tinnitus.	0.54
17. Tinnitus makes me feel tired.	0.58
18. Tinnitus makes me feel insecure.	0.63
19. Tinnitus contributes to a feeling of general ill health.	0.66
20. Tinnitus affects the quality of my relationships.	0.52
21. Tinnitus has caused a reduction in my speech understanding ability.	0.61
22. Tinnitus makes me feel annoyed.	0.57
23. Tinnitus interferes with my speech understanding when listening to the television.	0.62
24. Tinnitus makes me feel anxious.	0.70
25. I think I have a healthy outlook on tinnitus.	0.56
26. I have support from my friends regarding my tinnitus.	0.44
27. I feel frustrated frequently because of tinnitus.	0.65

Factor	Eigenvalue	% of Variance	Cumulative % of Variance
1	11.50	42.6	42.6
2	2.53	9.4	51.9
3	1.52	5.6	57.6

Table 2. Factor pattern matrix for the three factors and correlations among the three factors. The variables are grouped according to factor pattern.

Item	Pattern Matrix		
	Factor 1	Factor 2	Factor 3
13	0.87	-0.10	0.22
11	0.86	-0.04	-0.03
19	0.85	-0.07	-0.05
14	0.83	0.02	-0.05
24	0.80	0.08	-0.02
16	0.79	-0.14	-0.03
15	0.76	0.15	-0.03
27	0.74	0.08	0.14
17	0.74	0.05	-0.10
22	0.70	0.02	0.19
1	0.70	0.04	-0.10
9	0.66	0.15	-0.14
18	0.64	0.27	-0.14
12	0.54	-0.02	-0.25
20	0.48	0.36	-0.10
6	-0.16	0.89	0.01
21	-0.10	0.83	0.02
4	0.02	0.81	-0.01
3	0.02	0.75	0.12
23	0.10	0.72	0.06
7	0.23	0.66	0.01
10	0.41	0.41	-0.12
5	0.26	0.39	0.09
25	-0.26	-0.04	0.73
26	-0.03	0.12	0.64
8	0.31	0.07	0.52
2	0.25	0.12	0.36

	Correlation		
	Factor 1	Factor 2	Factor 3
Factor 1	1.00		
Factor 2	0.49	1.00	
Factor 3	0.10	0.16	1.00

Fifteen items loaded highly in Factor 1, eight items loaded highly on Factor 2, and only four items loaded highly on Factor 3. Examination of these items indicated that Factor 1 reflected the effects of tinnitus on the patients' social (items 1, 12, 15, 20), emotional (items 11, 13, 18, 22, 24, 27), and physical behaviors (items 9, 14, 16, 17, 19). Factor 2 included items that reflected the patients' hearing ability (items 3, 4, 5, 6, 21, 23). The only exceptions were items 7 ("I feel uneasy in social situations because of ") and 10 ("Tinnitus creates family problems"). Factor 3 included items 2, 8, 25, and 26. These items represented the patients' view of tinnitus—whether it had worsened (item 2), whether the outlook was healthy (item 26), whether help was available (item 27), and the patient's view of others' perception on tinnitus (item 8).

Reliability

A reliability analysis was performed to determine the Cronbach's α and the item-total correlation coefficients. A Cronbach's α of 0.94 was obtained for the 27-item questionnaire, indicating good internal consistency reliability. Item-total correlations ranged from 0.15 to 0.81. Table 3 summarizes the item-total correlation coefficients for the items.

Table 3. Item-total correlation coefficients for each item on the 27-item questionnaire ($n = 275$).

Item	Item-Total Correlation
1	0.64
2	0.36
3	0.55
4	0.59
5	0.51
6	0.47
7	0.67
8	0.38
9	0.67
10	0.64
11	0.75
12	0.50
13	0.75
14	0.76
15	0.81
16	0.60
17	0.68
18	0.74
19	0.70
20	0.67
21	0.49
22	0.67
23	0.61
24	0.78
25	0.15
26	0.15
27	0.74

The Cronbach's α changed when separate reliability analyses were performed on each factor scale to examine the reliability of each scale separately. Cronbach's α was 0.95 when computed for the items that loaded on Factor 1, 0.88 when computed for items that loaded on Factor 2, and 0.47 when computed for items that loaded on Factor 3. These results indicated poor reliability if the four items on the Factor 3 subscale were used alone. This low level of reliability is probably due to the small number of items on the scale and low correlations among these items.

Construct Validity

Table 4 summarizes patients' responses on each subscale of the tinnitus handicap questionnaire and on each validation measure. Responses were tallied according to the patients' gender. Multiple regression analysis, using the validation measure. Responses were tallied according to the patients' gender. Multiple regression analyses, using the validation measures and responses on the tinnitus questionnaire as dependent variables, were performed to determine if gender, the responses of all 53 patients were combined for analysis.

Patients reported a mean perceived tinnitus loudness of 46.7 (out of 100), and a mean total score of 32.4 on the tinnitus handicap questionnaire. The mean scores on the life satisfaction and depression scales were 2.98 and 2.00, respectively, indicating that these patients were satisfied with life or free from depression most of the time. The average patient had a score of 1.38 on the health status profile, and 2.96 instances of illness on the illness checklist, indicating that the average patient had few medical problems. A mean general

health rating of 3.49 indicated that patients perceived their general health as being the same or slightly better than it was 6 months ago or when compared to others. A mean social desirability rating of 18.0 suggested that the average tinnitus patient desired as much social approval as did the average individual with no tinnitus (Crowne & Marlowe, 1964).

Table 5 summarizes the correlations between scores on the tinnitus handicap questionnaire (and its three subscales) and the validation measures. The patients' mean threshold (in dB HL) from 250 to 8000 Hz at octave intervals was also used for correlation. Because some of the 53 patients did not complete all items on the validation measures, data on those patients were deleted in the analysis. The number of cases (or patients) and the probability (P) that the obtained correlation coefficient deviates from the true null hypothesis are also indicated.

The results in Table 5 indicate no correlation between scores on the tinnitus handicap questionnaire and the patients' age or their scores on the social desirability scale. Low correlation coefficients (0.2–0.5) were found between scores on the tinnitus handicap questionnaire and the loudness matching task, the minimal masking level, the health status profile, and the illness checklist. Correlations between the

total score on the tinnitus handicap questionnaire were moderately high with the perceived tinnitus loudness judgment ($r = 0.57$), life satisfaction ($r = -0.54$), mean hearing threshold ($r = 0.52$), depression ($r = 0.63$), and general health status ($r = 0.54$). Similar correlations were found between scores on the Factor 1 subscale (i.e., social, emotional, and physical behaviors) and these measures (except hearing threshold). Scores on the Factor 2 subscale (i.e., hearing) correlated moderately with perceived tinnitus loudness ($r = 0.45$) and highly with the mean hearing threshold ($r = 0.74$). Factor 3 (i.e., patient's view of tinnitus) correlated most highly with life satisfaction ($r = -0.62$). These correlations suggested that performance on the tinnitus handicap questionnaire (and its subscales) did reflect the hearing, emotion, physical, and social behavioral status of the tinnitus patients.

The high correlation between subjective judgments of tinnitus loudness and tinnitus handicap suggested that tinnitus loudness could be a factor in affecting its perceived handicap. However, the low correlation between tinnitus handicap and loudness matching ($r = 0.27$) and the low correlation between the subjectively (rating) and objectively determined tinnitus loudness (loudness match, $r = 0.32$; MML, $r = 0.26$) suggested that the subjective and objective measures may not be assess-

Table 4. Mean scores on each of the validation measures and on each subscale of the tinnitus questionnaire for the 53 subjects.

	Male Patients (<i>n</i> = 33)	SD	Female Patients (<i>n</i> = 20)	SD	All Patients (<i>n</i> = 53)	SD
Loudness match (in sonos)	13.0	24.7	14.5	29.2	13.1	28.0
MML (dB SL)	21.9	21.8	26.1	22.9	25.4	18.5
Perceived loudness (0–100)	43.7	21.6	51.2	27.2	46.7	24.0
Total	31.9	21.3	33.3	23.9	32.4	22.1
Factor 1	28.7	23.0	35.2	30.5	31.2	26.0
Factor 2	31.8	26.1	22.7	25.0	28.4	25.8
Factor 3	45.3	21.9	47.4	20.1	46.1	21.1
Life satisfaction	3.0	0.50	2.96	0.62	2.98	0.55
Depression	3.04	0.41	3.0	0.53	2.00	0.46
Health	1.38	0.29	1.37	0.16	1.38	0.25
General health	3.48	0.83	3.51	0.76	3.49	0.79
Illness checklist	3.03	2.24	2.85	1.95	2.96	2.12
Social desirability	18.8	6.6	17.5	6.3	18.00	6.60

Table 5. Correlation between scores on the tinnitus questionnaire (Total, Factor 1, Factor 2, Factor 3) and the validation measures. Included also are the number of cases and the probability that the observed correlation deviates from the null hypothesis.

	Age	Loudness Match (sonos)	MML for Noise	Perceived Loudness	Mean Hearing Threshold	Life Satisfaction	Depression	Health Status	General Health	No. of Illness	Social Desirability
Total	–0.03 <i>n</i> = 53 <i>p</i> = 0.83	0.27 <i>n</i> = 44 <i>p</i> = 0.04	–0.14 <i>n</i> = 46 <i>p</i> = 0.17	0.57 <i>n</i> = 45 <i>p</i> = 0.0001	0.52 <i>n</i> = 45 <i>p</i> = 0.0003	–0.54 <i>n</i> = 53 <i>p</i> = 0.0001	0.63 <i>n</i> = 53 <i>p</i> = 0.0001	–0.46 <i>n</i> = 53 <i>p</i> = 0.0006	–0.54 <i>n</i> = 51 <i>p</i> = 0.0001	0.37 <i>n</i> = 53 <i>p</i> = 0.006	–0.12 <i>n</i> = 53 <i>p</i> = 0.40
Factor 1	–0.03 <i>n</i> = 53 <i>p</i> = 0.83	0.21 <i>n</i> = 44 <i>p</i> = 0.08	–0.10 <i>n</i> = 46 <i>p</i> = 0.26	0.56 <i>n</i> = 45 <i>p</i> = 0.0001	0.33 <i>n</i> = 45 <i>p</i> = 0.03	–0.54 <i>n</i> = 53 <i>p</i> = 0.0001	0.63 <i>n</i> = 53 <i>p</i> = 0.0001	0.43 <i>n</i> = 53 <i>p</i> = 0.001	–0.55 <i>n</i> = 51 <i>p</i> = 0.0001	0.34 <i>n</i> = 53 <i>p</i> = 0.01	–0.14 <i>n</i> = 53 <i>p</i> = 0.31
Factor 2	–0.02 <i>n</i> = 53 <i>p</i> = 0.89	0.26 <i>n</i> = 44 <i>p</i> = 0.04	–0.17 <i>n</i> = 46 <i>p</i> = 0.13	0.45 <i>n</i> = 45 <i>p</i> = 0.002	0.74 <i>n</i> = 45 <i>p</i> = 0.0001	–0.26 <i>n</i> = 53 <i>p</i> = 0.06	0.37 <i>n</i> = 53 <i>p</i> = 0.007	0.30 <i>n</i> = 53 <i>p</i> = 0.03	–0.32 <i>n</i> = 51 <i>p</i> = 0.02	0.33 <i>n</i> = 53 <i>p</i> = 0.02	–0.05 <i>n</i> = 53 <i>p</i> = 0.72
Factor 3	–0.03 <i>n</i> = 53 <i>p</i> = 0.85	0.27 <i>n</i> = 44 <i>p</i> = 0.04	–0.13 <i>n</i> = 46 <i>p</i> = 0.19	0.40 <i>n</i> = 45 <i>p</i> = 0.007	0.29 <i>n</i> = 45 <i>p</i> = 0.05	–0.62 <i>n</i> = 53 <i>p</i> = 0.0001	0.61 <i>n</i> = 53 <i>p</i> = 0.0001	0.45 <i>n</i> = 53 <i>p</i> = 0.0007	–0.45 <i>n</i> = 51 <i>p</i> = 0.001	0.22 <i>n</i> = 53 <i>p</i> = 0.114	–0.03 <i>n</i> = 53 <i>p</i> = 0.82

ing the same attribute. There are several factors that may complicate the understanding of the relationship between "subjective" and "objective" loudness estimates. Subjective estimates of loudness may actually include the patient's reaction to the tinnitus loudness; for example, stress and annoyance. Therefore, it may correlate higher with their overall perception of tinnitus handicap than objective tinnitus loudness measures. The difficulty of quantifying tinnitus loudness has been discussed by other investigators (Goodwin & Johnson, 1980; Jakes, Hallam, Chambers, & Hinchliffe, 1985; Penner, 1984; Reed, 1960; Tyler & Conrad-Armes, 1983).

Multiple regression analyses were performed with each of the psychological and psychophysical measures as the dependent variable and scores on the different tinnitus subscales as the independent variables. Table 6 presents the standardized β -weights, the raw B-weights, the proportion of variance explained by the three factors (i.e., R^2), and the F -values for each regression equation.

Table 6 shows that Factor 1 was the only variable that contributed significantly ($p < 0.05$) to predicting scores on the perceived tinnitus loudness and general health status measures, whereas Factor 2 was the only variable that contributed significantly to predicting scores on the mean hearing loss scale. This suggests that patients' perceived hearing handicap may not be due to their tinnitus per se, but may have resulted partially from their hearing insensitivity. Both Factor 1 and Factor 3 contributed significantly to predicting life satisfaction and depression. None of the three factors contributed significantly to predicting responses on the loudness matching task, the MML task, symptom report, functional status, the number of illnesses, or social desirability.

Cumulative Percent Distribution of Ratings

Responses on the tinnitus handicap questionnaire represented a wide range of handicaps due to tinnitus. Figure 2 shows the cumulative percent distribution of responses on the different subscales from the tinnitus questionnaire. This distribution was based on the responses of 275 patients.

Figure 2 shows the proportion of patients who received a particular score. For example, let us assume that patient A received a mean total score of 30% on the tinnitus questionnaire. This corresponds to approximately the 50th percentile, suggesting that approximately half of the 275 patients in this sample scored less than 30% on the questionnaire.

DISCUSSION

The results of this study indicated that the 27-item tinnitus handicap questionnaire has high internal consistency reliability (Cronbach's $\alpha = 0.94$) and good construct validity as reflected by its moderately high correlation with measures of life satisfaction, depression, general health status, and perceived tinnitus loudness. The 27-item scale, like the Tinnitus Severity Scale (Sweetow & Levy, 1990), may be used to measure the tinnitus patients' perception of their physical and psychological well-being.

There are three factors underlying patients' responses

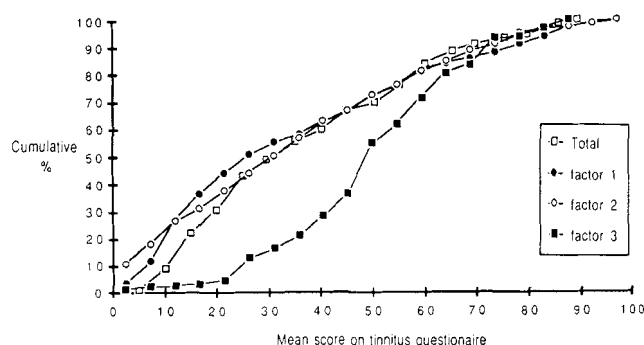


Figure 2. Cumulative percent-mean score functions for total and subscale mean scores ($N = 275$).

Table 6. Summary of regression analyses on each validation measure.

Dependent Variables	Raw B-Weights			Standardized β -Weights			R^2	F -Value
	Factor 1	Factor 2	Factor 3	Factor 1	Factor 2	Factor 3		
Loudness match ($n = 44$)	-0.048 $p = 0.84$	0.211 $p = 0.26$	0.288 $p = 0.31$	-0.044	0.198	0.220	0.10	1.51 $p = 0.23$
MML for noise ($n = 46$)	0.043 $p = 0.80$	-0.116 $p = 0.38$	-0.097 $p = 0.60$	0.057	-0.155	-0.111	0.04	0.52 $p = 0.67$
Perceived loudness ($n = 45$)	0.372* $p = 0.03$	0.147 $p = 0.37$	0.056 $p = 0.77$	0.431*	0.151	0.050	0.33	6.71 $p = 0.0009$
Hearing loss ($n = 45$)	-0.634 $p = 0.36$	3.495* $p = 0.000$	0.365 $p = 0.65$	-0.143	0.794*	0.066	0.56	17.62 $p = 0.0000$
Life satisfaction ($n = 53$)	-0.006 $p = 0.06$	0.002 $p = 0.43$	-0.012* $p = 0.002$	-0.300	0.107	-0.470*	0.43	12.10 $p = 0.0000$
Depression ($n = 53$)	0.007* $p = 0.008$	-0.0003 $p = 0.89$	0.008* $p = 0.01$	0.417*	-0.018	0.350*	0.47	14.50 $p = 0.0000$
Health ($n = 53$)	0.002 $p = 0.25$	0.0006 $p = 0.68$	0.003 $p = 0.08$	0.214	0.062	0.288	0.24	5.20 $p = 0.003$
General health ($n = 51$)	-0.014* $p = .01$	0.0002 $p = 0.96$	-0.005 $p = 0.40$	-0.464*	0.007	-0.138	0.32	7.23 $p = 0.0004$
Illness ($n = 53$)	0.020 $p = 0.22$	0.016 $p = 0.23$	-0.002 $p = 0.93$	0.242	0.195	-0.016	0.14	2.74 $p = 0.053$
Social desirability ($n = 53$)	-0.002 $p = 0.27$	0.0003 $p = 0.81$	0.0009 $p = 0.59$	-0.232	0.042	0.100	0.03	0.47 $p = 0.71$

* Denotes significance at $p < 0.05$ level.

on the 27-item questionnaire. Factor 1 addresses the individuals' physical health (items 9, 14, 16, 17, 19), emotional status (items 11, 13, 18, 22, 24, 27), and the social consequences of tinnitus (items 1, 12, 15, 20). Factor 2 addresses the individuals' hearing difficulty (items 3, 4, 5, 6, 21, 23) related to tinnitus. Factor 3 addresses the patients' view on tinnitus (items 2, 8, 25, 26).

The usefulness of a test (or its subscale) depends on its validity and reliability. Although Factor 3 subscale correlated moderately with life satisfaction and depression scales and may have predictive value for therapeutic success, its low internal consistency reliability (Cronbach's $\alpha = 0.47$) suggests that its usefulness as an independent subscale is questionable. Alternative measures may need to be used or developed that address the patients' view on tinnitus. For example, the Social Provisions Scale (Cutrona, 1984) which measures an individual's social relationship could replace items 8, 25, and 26 on the Factor 3 subscale. This scale could be used in conjunction with the tinnitus handicap questionnaire to reliably measure patients' handicap and perception of social support.

Clinical Application

It needs to be stressed that the present study was normed on individuals who reported tinnitus during their audiological/otological evaluation. The responses from these patients would likely cover a wide range of handicaps and would represent the most typical population of patients seen in audiological/otological facilities. For this reason, the correlation seen between the tinnitus questionnaire and the validation measures may be different if another group of tinnitus patients, for example, those whose tinnitus was the primary complaint, was used. For this group of patients, they would likely score high on the tinnitus handicap questionnaire. In addition, they would likely show different correlations with the validation measures. However, because the usefulness of a standardized questionnaire is measured by how representative it is of the population that it intends to measure, we feel that our choice of patients to standardize this questionnaire is appropriate.

Because the tinnitus handicap questionnaire is standardized on 27 items, it is important that clinicians administer the full 27-item questionnaire to their tinnitus patients to ensure a valid comparison between a patient's score and the normative data. It is also important that the clinician invert the responses obtained on items 25 and 26 by subtracting them from 100 before calculating the average score on each factor scale. The mean scores on Factor 1 and Factor 2 subscales can be compared with the normative data. However, for reasons discussed previously, we feel that the Factor 3 subscale score cannot be considered reliable.

The mean score on the tinnitus questionnaire reflects the degree of handicap due to tinnitus. The severity of the patient's handicap can be examined in two ways.

One is to examine the absolute magnitude of the handicap. In general, the larger the score, the more severe the handicap. The other alternative is to compare the patient's mean score on each subscale with other patients' scores (i.e., norms) presented in Figure 2. For example, let us assume that a patient obtained a mean score of 75% on the Factor 1 subscale, and 25% on the Factor 2 subscale. Examination of Figure 2 shows that a score of 75% on the Factor 1 subscale places the patient's handicap at the 85th percentile, while a score of 25% places the patient's hearing difficulties at the 45th percentile. This suggests that the patient's perceived social, emotional and physical handicaps due to tinnitus are more severe than 85% of the patients who comprised the normative sample and that his/her hearing difficulties are more severe than 45% of the normative sample.

Unfortunately, being able to determine the percentile rank of the patient's tinnitus handicap does not mean that one can easily set a criterion score to determine the need for therapeutic intervention. Such a decision can only be made based on all aspects of the patient's well being. Factors such as the patient's medical history may affect one's decision to enroll the patient into a medication program; support from the patient's family may affect the outcome of any psychological treatments. Nonetheless, scores on the tinnitus handicap questionnaire can supply an important piece of diagnostic information for the clinician to determine their perceived handicap relative to other tinnitus patients. This may allow one to set priority for patient intervention. However, it cannot and should not be the only criterion that decides on the need for a particular type of intervention program.

Scores on the Factor 1 and Factor 2 subscales may provide insights for remediation of tinnitus patients. Because of the moderate correlation between Factor 1 and Factor 2 scores ($r = -0.46$), a high score on both subscales suggests that the patient's perceived handicap may be affected by hearing sensitivity. Remediation programs may include the use of appropriate amplification devices in addition to tinnitus treatment (e.g., counseling). In addition, items within a subscale that had a high score may indicate the specific problem areas experienced by the patient. Knowing this information may help guide the clinician in structuring the remediation program.

A high Factor 1 score and a low Factor 2 score suggests that the patient's difficulty may be primarily in the social, emotional, and physical aspects of his/her well-being. It may suggest that the patients have minimal difficulties with hearing, or that they can cope with their hearing difficulties effectively. Remediation programs should be directed at the handicap areas. However, the use of appropriate amplification in the presence of elevated thresholds may be beneficial also.

Patients with low Factor 2 scores may exhibit only a mild degree of tinnitus (i.e., low Factor 1 score). For these patients, one may simply provide necessary coun-

selling. Patients with low Factor 1 score and high Factor 2 score may only need intervention for amplification.

The high internal consistency reliability of the tinnitus questionnaire and its moderate construct validity suggest that it may be used to monitor a patient's progress during a treatment program, or to compare the effectiveness of different tinnitus treatment programs. Additional data are necessary to evaluate these possibilities.

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APPENDIX. Mean score, item-total correlation for each item, percent of time (%) subjects assigned a "0" or "100" rating for each item, and the final 27 items that were selected from the 87-item preliminary questionnaire.

Item	Description	Mean Score	Item-Total Correlation Coefficient	% of Subjects Showing Extreme Ratings		Is Item Selected for Final Questionnaire (X)
				0	100	
HEAR 1:	Tinnitus interferes with my speech understanding when talking with someone in a quiet room.	26.3	0.64	40	5	
HEAR 2:	Tinnitus interferes with my speech understanding when talking with someone in a noisy room.	35.2	0.31	18	20	X
HEAR 3:	Tinnitus interferes with my speech understanding when listening to the television.	36.6	0.61	23	7	X
HEAR 4:	Tinnitus interferes with my ability to tell where sounds are coming from.	43.8	0.43	28	16	X
HEAR 5:	Tinnitus interferes with my ability to recognize sounds like door bells and telephone ringing.	29.5	0.45	39	7	
HEAR 6:	Tinnitus interferes with my understanding of children's speech.	31.1	0.53	34	4	
HEAR 7:	I am unable to follow a conversation during meetings because of tinnitus.	34.3	0.74	33	7	X
HEAR 8:	Tinnitus interferes with my recognition of warning signals.	19.5	0.70	48	1	
HEAR 9:	Tinnitus has caused a reduction in my speech understanding ability.	36.4	0.54	32	8	X
HEAR 10:	Tinnitus interferes with listening to the radio.	29.7	0.64	38	7	
HEAR 11:	Tinnitus interferes with my use of the telephone.	31.3	0.59	33	4	
HEAR 12:	I am hesitant to respond to what is said to me because of tinnitus.	40.2	0.56	22	9	
LIFE 1:	Tinnitus interferes with my appreciation of music.	28.0	0.63	44	5	
LIFE 2:	Tinnitus causes me to avoid noisy situations.	47.3	0.43	34	19	X
LIFE 3:	Tinnitus interferes with my ability to do mental work.	28.9	0.71	43	10	
LIFE 4:	Tinnitus interferes with my ability to do physical work.	10.1	0.38	74	2	
LIFE 5:	Tinnitus creates family problems.	24.3	0.68	46	5	X
LIFE 6:	Tinnitus causes me to withdraw and avoid friends.	16.9	0.61	58	4	
LIFE 7:	I am unable to relax because of tinnitus.	28.1	0.71	41	11	X
LIFE 8:	I am less effective at my job when I have tinnitus.	26.0	0.65	43	10	
LIFE 9:	I socialize less because of tinnitus.	22.2	0.65	57	8	
LIFE 10:	Tinnitus causes me to be easily distracted from my work.	22.3	0.61	52	4	
LIFE 11:	I feel uneasy in social situations because of tinnitus.	28.7	0.67	41	8	X
LIFE 12:	Tinnitus affects the quality of my relationships.	27.4	0.78	45	6	X

APPENDIX CONT.

Item	Description	Mean Score	Item-Total Correlation Coefficient	% of Subjects Showing Extreme Ratings		Is Item Selected for Final Questionnaire (X)
				0	100	
LIFE 13:	I cannot concentrate because of tinnitus.	28.0	0.67	42	9	X
LIFE 14:	Tinnitus discourages me from going to movies.	17.0	0.73	65	6	
LIFE 15:	Tinnitus discourages me from attending social events.	21.9	0.73	59	6	
LIFE 16:	Tinnitus interferes when I am reading	26.7	0.60	43	6	
LIFE 17:	My relationship with my friends suffered because of tinnitus.	19.6	0.73	59	3	
LIFE 18:	I do not interact with children because of tinnitus.	12.9	0.47	70	0	
LIFE 19:	I do not take trips because of tinnitus.	9.3	0.44	76	2	
LIFE 20:	I abandoned my favorite pastimes because of tinnitus.	13.3	0.61	72	4	
LIFE 21:	I am less outgoing because of tinnitus.	19.4	0.70	57	4	
LIFE 22:	Tinnitus interferes with my driving.	10.1	0.69	71	1	
LIFE 23:	I find it difficult to explain what tinnitus is to others.	37.3	0.50	43	19	X
LIFE 24:	I do not enjoy life because of tinnitus.	20.6	0.71	58	7	X
LIFE 25:	I do not enjoy my work because of tinnitus.	16.4	0.72	65	3	
LIFE 26:	I complain more because of tinnitus.	22.8	0.70	50	6	X
LIFE 27:	My tinnitus is better when I concentrate on my work.	48.0	0.15	31	21	
HEALTH 1:	I have trouble falling asleep at night because of tinnitus.	43.9	0.33	29	23	X
HEALTH 2:	Tinnitus wakes me up in the morning.	23.4	0.51	54	8	
HEALTH 3:	Tinnitus is worse in the morning.	17.5	0.51	57	3	
HEALTH 4:	I am dependent upon drugs for relief of tinnitus.	6.4	0.35	84	2	
HEALTH 5:	Tinnitus gives me a headache.	14.7	0.62	62	6	
HEALTH 6:	Tinnitus is painful.	11.5	0.60	62	6	
HEALTH 7:	Tinnitus makes me dizzy.	24.8	0.49	58	7	
HEALTH 8:	Tinnitus makes me feel tired.	21.2	0.75	57	5	X
HEALTH 9:	Tinnitus contributes to a feeling of general ill health.	23.6	0.70	51	8	X
HEALTH 10:	Tinnitus causes stress.	42.8	0.69	17	14	X
HEALTH 11:	I felt lethargic because of tinnitus.	21.3	0.63	49	5	
HEALTH 12:	Tinnitus causes me to lose sleep.	32.9	0.46	39	17	
HEALTH 13:	Tinnitus makes me nervous.	40.6	0.67	31	18	
HEALTH 14:	I believe that tinnitus is a symptom of a serious illness.	12.9	0.23	59	4	
HEALTH 15:	My tinnitus has gotten worse over the years.	54.3	0.32	23	30	X
HEALTH 16:	The increased severity of tinnitus worries me.	50.8	0.53	28	26	
HEALTH 17:	I often receive relief for tinnitus.	15.2	-0.07	70	5	
HEALTH 18:	I think tinnitus therapy is a waste of time and money.	16.0	0.06	57	3	
EMOTION 1:	Tinnitus causes me to feel depressed.	31.0	0.71	42	10	X

APPENDIX CONT.

Item	Description	Mean Score	Item-Total Correlation Coefficient	% of Subjects Showing Extreme Ratings		Is Item Selected for Final Questionnaire (X)
				0	100	
EMOTION 2:	I feel frustrated frequently because of tinnitus.	41.0	0.67	28	12	X
EMOTION 3:	Tinnitus makes me feel annoyed.	44.4	0.55	19	14	X
EMOTION 4:	Tinnitus makes me feel insecure.	24.3	0.71	52	5	X
EMOTION 5:	I feel disoriented because of my tinnitus.	25.3	0.74	51	4	
EMOTION 6:	I am more easily confused when I have tinnitus.	27.3	0.71	47	8	
EMOTION 7:	Tinnitus makes me feel anxious.	30.3	0.78	40	10	X
EMOTION 8:	I feel isolated from others because of tinnitus.	21.3	0.76	55	5	
EMOTION 9:	Tinnitus causes me to feel apprehensive.	31.4	0.49	45	8	
EMOTION 10:	I am easily agitated because of tinnitus.	28.0	0.64	43	9	
EMOTION 11:	I feel tinnitus lowers my self-esteem.	17.6	0.57	59	5	
EMOTION 12:	I think my tinnitus can never be helped.	31.1	0.07	45	14	
EMOTION 13:	I think I have a healthy outlook on tinnitus.	64.5	-0.10	17	31	X
EMOTION 14:	I think my tinnitus can be helped one day.	58.8	0.08	17	28	
EMOTION 15:	I often feel lonely because of my tinnitus.	18.0	0.75	62	5	
EMOTION 16:	Tinnitus lowered my self-confidence	22.5	0.66	54	7	
EMOTION 17:	I have considered suicide because of tinnitus.	4.7	0.37	84	2	
EMOTION 18:	Tinnitus made me a less desirable person.	14.3	0.63	68	4	
EMOTION 19:	My outlook on life is terrible because of the tinnitus.	11.8	0.64	74	2	
EMOTION 20:	I feel irritated because of my tinnitus.	36.1	0.57	31	13	
OTHER 1:	No one but a tinnitus sufferer can understand tinnitus.	50.6	0.33	23	27	
OTHER 2:	People stay away from me because of my tinnitus.	6.0	0.25	81	0	
OTHER 3:	I have support from my friends regarding my tinnitus.	64.1	0.23	34	25	X
OTHER 4:	The general public does not know about the devastating nature of tinnitus.	64.1	0.23	13	33	X
OTHER 5:	People patronize me because of my tinnitus.	9.9	0.44	74	1	
OTHER 6:	No one understands the anxiety I am going through because of tinnitus.	38.8	0.37	33	17	
OTHER 7:	People respect me less because I have tinnitus.	5.7	0.32	81	1	
OTHER 8:	Tinnitus is worse in quiet.	70.0	0.16	18	50	
OTHER 9:	Tinnitus is worse in noise.	28.4	0.34	51	13	
OTHER 10:	My tinnitus has gotten better over the years.	13.9	-0.05	75	6	